CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

December 2012

CL 404 Water Resources Engineering & Management

Date: 07.12.2012, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Explain the process of river meandering with neat sketch. [2]

(b) Discuss the importance of river training. What are different types of river training works? Discuss **any Two** of them in detail. [5]

Q - 2 (a) What are the objectives of water resources development? Classify WRD Projects with their functional requirements. How any WRDP is formulated? [6]

(b) Discuss the process of water resources project evaluation in detail. [5]

(c) What are physical factors affecting WRDP? [3]

OR

Q - 2 (a) Discuss planning strategy for water resources development projects. List out various steps involved to plan WRP. Explain in detail data required to be collected for planning of a WRP. [6]

(b) Explain the environmental aspects of water resources project planning. [4]

(c) Write a short note on water conservation measures. [4]

Q - 3 (a) Define drought and flood. What are the causes of both? Enlist flood management activities. Briefly discuss the structural measures for flood mitigation. [7]

(b) What are merits and demerits of hydro-power as compared to other power sources? Also explain terms load factor and plant factor in context of hydro-power plant. [4]

(c) Discuss types of hydro-power based on operating head with neat sketches. [3]

OR

Q - 3 (a) What is conservation of water? Discuss the importance of water conservation in India. Discuss in detail the water management activities to tackle drought effects. [7]

(b) Give detailed classification of hydro-power scheme. Show different components of a hydro-power plant with neat sketch. [4]

(c) A hydel station having a base load of 23 MW capacity. The yearly output is 100×10^6 kWh. The peak load taken by station is 15 MW. Calculate annual load factor, plant use factor and capacity factor for the plant. [3]

SECTION – II

Q - 4 Following data has been collected during contour survey of a reservoir site: [7]

R.L. (m)	Area (ha.)
150.00	18
175.00	27
200.00	40

The capacity at datum is observed 10 ha m. Develop the equations for Area-elevation curve and Capacity-elevation curve for given data. Also calculate the capacity at RL175.00 m & RL 200.00 m.

Q - 5 (a) Define reservoir. Classify reservoirs by different ways. Explain factors affecting selection of site for a reservoir. [4]

(b) How the investigation for reservoir planning is carried out? Explain in detail. [4]

(c) The daily flows in a river for two years are listed in the table given below. Compute the annual and daily flow for 63% dependability of river flow. [6]

Sr. No.	Mean Value of Daily Discharge, (m^3/s)	Year 2000	Year 2001
		No. of days the flow	No. of days the flow
1	259.50	4	2
2	214.45	2	3
3	207.45	1	2
4	156.45	2	3
5	102.45	27	40
6	83.95	24	35
7	67.45	40	55
8	56.95	54	22
9	35.95	63	60
10	16.45	22	26
11	9.45	54	65
12	4.45	72	52

OR

Q - 5 (a) Explain storage zones of reservoir with neat sketch. [4]

(b) The yearly data for a catchment of a proposed reservoir site for 35 years is given in following table. Compute the dependable rainfall for 54% & 72% dependability. [4]

Year	1900	1901	1902	1903	1904	1905	1906	1907	1908	1909
Rainfall (cm)	104	92	95	87	69	91	128	120	111	110
Year	1910	1911	1912	1913	1914	1915	1916	1917	1918	1919
Rainfall (cm)	83	82	94	128	101	112	80	85	116	120
Year	1920	1921	1922	1923	1924	1925	1926	1927	1928	1929
Rainfall (cm)	131	138	78	76	90	113	126	70	118	140
Year	1930	1931	1932	1933	1934	-	-	-	-	-
Rainfall (cm)	52	59	72	102	79	-	-	-	-	-

(c) The following table gives the mean monthly flows in a river during year 1990. Calculate the minimum storage required to maintain a demand rate of 90 m^3/sec for one month by analytical method. [6]

Month	Jan	Feb	March	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Mean Flow (m^3/sec)	20	60	200	300	200	150	100	80	60	40	30	25

Q - 6 (a) Explain reservoir losses in detail. [3]

(b) Explain the following terms: [3]

(i) Density Currents (ii) Trap Efficiency (iii) Capacity Inflow Ratio

(c) The following table gives the monthly inflow, contemplated demand from a proposed reservoir, mean monthly evaporation and rainfall. Estimate the minimum storage that is necessary to meet the demand. If the average reservoir area is assumed as 400 ha. Prior rights make it obligatory to release the full natural flow **or** 15ha.m/month, whichever is minimum, Assume, only 28 % of the rainfall on land area to be flooded by reservoir has reached the stream in the past. [8]

Month	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Monthly Inflow (ha. m)	1.2	0	0	0	0	0	240	480	1.0	0.6	0.5	0.2
Monthly Demand (ha. m)	16	15	10	5	4	4	5	5	10	16	17	17
Evaporation (cm)	1.26	1.26	1.82	7.14	10.78	1.12	7.56	8.19	7.56	6.72	5.46	1.4
Rainfall (cm)	1.3	2	0.6	0	0	1.1	16	16	2	0.8	0	0

OR

Q - 6 (a) Write a short note on reservoir clearance. [3]

(b) What measures should be taken to control the sedimentation before and after constructing the reservoir? [6]

(c) A reservoir has a capacity of 540 ha. m and catchment area 360 km^2. The annual streamflow averages 11cm of runoff and production of annual sediment is 0.036ha.m/km^2. Calculate, annual flood inflow and sediment inflow in the reservoir in Mm^3. [5]

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